

Root Cause Analysis Guide

E-Bike Manufacturing — Enhanced MES Reference

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A structured reference for investigating quality and reliability problems in the MES. It links the recorded root-cause categories, downtime reasons and predictive-maintenance failure modes to concrete investigation steps and corrective / preventive actions.

1. RCA Workflow

- 1 **Trigger** — a 'fail' QC result, a severity-4/5 defect, an unplanned downtime event, or a high-priority predictive alert.
- 2 **Contain** — quarantine affected units, tag the machine, protect traceability (lot numbers).
- 3 **Investigate** — gather data from Defects, Downtimes, OEE Metrics and Vibration Analysis for the machine/order.
- 4 **Identify root cause** — classify into one of the six categories using 5-Whys / fishbone.
- 5 **Correct & prevent** — apply the disposition and a preventive action; record in ActionTaken.
- 6 **Verify** — confirm the defect rate returns to baseline over the next lots; close the record.

2. By Root-Cause Category

Process Variation (276 findings)

Drift within nominal limits producing borderline output.

- Pull OEE Metrics and Vibration Analysis for the run window.
- Chart the parameter (SPC) to see the trend vs limits.
- Corrective: re-center the process; Process Adjusted disposition.
- Preventive: narrow control limits; add in-line monitoring.

Machine Calibration (264 findings)

Equipment drifted out of calibration.

- Check LastMaintenanceDate vs NextMaintenanceDate on the machine.
- Review recent OEE Performance/Quality trend for the asset.
- Corrective: recalibrate, re-run first-piece verification.
- Preventive: tighten calibration frequency; SPC on the affected parameter.

Material Defect (262 findings)

Incoming component non-conformance.

- Trace the component lot via MaterialConsumption.LotNumber.
- Check the supplier ReliabilityScore and recent return history.
- Corrective: Returned to Supplier; quarantine remaining lot.
- Preventive: tighten incoming inspection; supplier corrective-action request.

Design Issue (260 findings)

Root cause in product or fixture design.

- Confirm the defect recurs across operators/machines/lots.
- Review BOM and fixture drawings for the feature.
- Corrective: engineering change request; interim concession if safe.
- Preventive: design review; update BOM/fixtures.

Operator Error (245 findings)

Assembly, handling or setup mistakes.

- Verify the operator was trained/certified for the task (Employees.Skills).
- Check work instructions were current and followed.
- Corrective: retrain, add poka-yoke or visual guide, adjust station layout.
- Preventive: skills-based work-order assignment; first-piece sign-off.

Tooling Wear (242 findings)

Worn tips, electrodes, fixtures.

- Check tool-life counter against replacement interval.
- Inspect fixtures/clamps for wear or play.
- Corrective: replace tooling, re-verify dimensions.
- Preventive: tool-life tracking with automatic replacement prompts.

3. Downtime Reason Reference

Investigate recurring downtime by reason. Planned reasons are optimised (reduce/consolidate); unplanned reasons drive RCA.

Reason	Category	Events	Avg (min)
Cleaning	planned	7	41
Meeting	planned	5	91
Setup/Changeover	planned	5	29
Shift Change	planned	3	88
Training	planned	3	75
Scheduled Maintenance	planned	2	383
Power Outage	unplanned	8	82
Quality Issue	unplanned	6	69
Material Shortage	unplanned	4	88
Operator Absence	unplanned	4	80
Equipment Failure	unplanned	3	153
Tool Breakage	unplanned	1	107

4. Predictive Failure Modes → Actions

The predictive-maintenance model raises eight failure modes. Each maps to a standard recommended action; treat these as the first response when an alert is confirmed (not a false positive).

Failure Mode	Alerts	Avg RUL (h)	Avg Conf.	Recommended Action
belt wear	16	344	0.75	Replace drive belts
lubrication failure	14	650	0.80	Check and replenish lubrication system
tool wear	12	533	0.79	Schedule tool replacement
spindle imbalance	12	738	0.76	Balance spindle assembly
bearing failure	12	697	0.80	Replace bearings during next maintenance window
motor fault	11	734	0.74	Inspect motor windings and connections
vibration excessive	10	583	0.75	Check machine mounting and alignment
overheating	6	669	0.73	Inspect cooling system and ventilation

See the *Predictive Maintenance & Condition Monitoring* guide for the alert triage workflow and the vibration signatures behind these modes.